

VII. EVALUATING RELEASE CONTROLS

A. Water Controls

1. Gravity Separation

Gravity separation is widely used as a waste treatment process for the removal of settleable solids, oils, grease and other material from water. Clarifiers, API separators, and inclined plate settlers are commonly used for gravity separation.

For this situation it is necessary to estimate the amount of water used in cleaning to calculate the amount of chemical dissolved in the water and not readily separated. The amount of water should be estimated as equivalent to 10 to 100 percent of batch volume (all material in the batch, not just the chemical under study), using the higher range for high viscosity materials in multi-component systems (i.e., several pieces of equipment). The amount of chemical lost in water is then the volume of water times the water solubility of the material. Material not dissolved (i.e., amount available minus amount dissolved) should be assumed to be incinerated or landfilled after separation. This estimation method ignores variation of chemical solubility in water, which may vary with temperature, pH, and the presence of dissolved salts and minerals. In addition, the method is based on the assumption that 100 percent of the suspended solids are removed during filtration or settling, resulting in a release of dissolved solids only. For liquid/liquid systems, the method does not consider formation of emulsions. Although this method does not describe actual systems, it provides an order of magnitude estimate in the absence of specific data.

This approach should be used whenever immiscible materials are washed from equipment and subsequently removed from rinsewater prior to wastewater treatment or discharge (other than in a large holding pond in which dilution may be assumed sufficient to dissolve most of the chemical).

2. Carbon adsorption

Carbon adsorption is a physical separation process in which organic and inorganic materials are removed from wastewater by the attraction and

accumulation of impurities on the surface of activated carbon. This method is typically used when a material that is present in concentrations of less than 10 percent is to be removed from a water stream that is otherwise relatively clean and free of suspended matter. Activated carbon is either added directly to the wastewater stream and subsequently filtered out or it is placed in a column through which the wastewater passes. Carbon adsorption can be used as a wastewater treatment method or for recovering valuable organics and inorganics from water streams. In either case, a portion of the chemicals is likely to remain in wastewater effluent that may be released to a stream or POTW.

Organic chemicals differ widely in their treatability by carbon adsorption. A survey performed by Troxler, Parmlee, and Barton (CEB 1984a) resulted in carbon adsorption removal efficiencies that varied from 42 to 99 percent for a variety of organic chemicals. This report, which is included in Volume II, presents results of the survey for chemicals that have been treated on an experimental level with carbon adsorption and for chemicals removed in full-scale treatment operations. The efficiencies presented in the tables may be used to estimate water releases of the chemicals that are included in the tables; however, for chemicals not included in the survey (e.g., new chemicals), a different release estimation method must be used.

To estimate water releases of chemicals not included in the survey, the chemical's treatability must first be estimated. Some properties used to assess a chemical's treatability are molecular weight, structure, polarity, solubility, and water conditions (pH and temperature). If adequate information is available, an assessment of the effect of these properties and conditions on a chemical's treatability must be made by the CEB engineer. CEB 1984a discusses the effect these parameters have on treatability. This document may be used to make reasonable worst case release estimates for new chemicals. If a judgment about a new chemical's treatability cannot be made from available information, data found in CEB 1984a for a similar existing chemical should be used.

B. Air Controls

Air pollutants entering an air control device can undergo one or more of the following: 1) they can be transferred from the air stream to another medium, 2) they can be modified to a less toxic state, 3) they can be destroyed through combustion or dissociation, or 4) they can pass through untreated. The physical characteristics of the pollutant to be removed generally determine which type of control device is used.

Estimates of releases to air must consider the control equipment efficiency. This efficiency should be based on the amount of pollutant removed from the air inlet stream of the control device by destruction, modification, or transfer to another medium.

$$\text{Percent efficiency} = \frac{X_{\text{inlet}} - X_{\text{outlet}}}{X_{\text{inlet}}} \times 100 \quad \text{Equation 7-1}$$

where: X_{inlet} = Total mass of pollutant X flowing to the air inlet of the control device in a given year
 X_{outlet} = Total mass of pollutant X flowing from the air outlet of the control device in a given year

The amount of pollutant transferred to and subsequently released in another medium (solid or water) must be included in the releases of that particular pollutant in that medium.

The best basis for an efficiency estimate is a measurement or test, a mass balance calculation, or a combination of measurement and mass balance calculations. If such data are not available, comparison of "controlled" and "uncontrolled" emission factors for the pollutant (chemical) of concern, engineering calculations, data on the operation parameters of the control device, or vendor data or guarantees that reflect actual operating conditions may be used. It is important to use data that reflect efficiency obtained during typical operations, not the theoretical optimum efficiency.

In the absence of typical operating data, treatment efficiency data cited in the open literature for a similar process may be used as an approximate guide. Without actual source test data for a specific emission stream and control system, the

removal efficiency can be assumed to equal total VOC removal efficiency if the chemical is volatile organic compound (not a particulate, metal, PCB, etc.).

Typical types of air release controls include incineration, adsorption, absorption, condensation, cyclones/mechanical collectors, fabric filters, electrostatic precipitators, and wet scrubbers.

1. Incineration

Incineration is used to control air release of chemicals that can be oxidized. Incineration is also used to reduce liquid and solid waste generated in a process. A discussion of incineration as a control for both air streams and waste streams is presented in Section VII.C.1.

2. Adsorption

In an adsorption process, a pollutant is adsorbed on the surface of the adsorbent until its capacity is reached. Common adsorbent materials used are activated carbon, resins, and molecular sieve materials. The adsorbent can then be regenerated. The pollutant is released in a more concentrated form, which is recovered or treated by further processing. The particular adsorption/regeneration process and the pollutant and its associated process parameters determine further processing steps. These can include incineration or condensation and decantation so the chemical can be recovered for recycling or disposal. Although adsorption is effective in the removal of various toxic chemicals from air, the regeneration and further processing steps may transfer some toxic substance to water or to solid waste streams, which must be considered releases to these media. Typically, the adsorption capacity increases with the molecular weight of the VOC being adsorbed. In addition, unsaturated compounds are generally more completely adsorbed than saturated compounds, and cyclical compounds are more easily adsorbed than linearly structured materials. Also, the adsorption capacity is enhanced by lower operating temperatures and higher concentrations. The VOCs characterized by low vapor pressures are more easily adsorbed than those with high vapor pressures.

In the absence of process specific data, carbon adsorption can be assumed to have maximum DREs in excess of 50 percent for VOC inlet concentrations over 200 ppmv, over 95 percent for VOC inlet concentrations over 1,000 ppm, and over 99 percent for VOC inlet concentrations over 5,000 ppmv (USEPA 1984). Adsorption can be used for VOC recovery for relatively small industrial VOC sources. Commercial adsorption systems are available for small flow rates (several hundred to tens of thousands cfm) and low VOC concentrations (usually several hundred up to several thousand ppm) (Spivey 1986).

3. Absorption

Absorption as a method of treating an emission is a physical or chemical process that transfers a components from a gas stream to a liquid. Although often used to recover products or raw materials, absorption also can serve as an emission control device. In this capacity, absorption has been used to control alcohols, acids, chlorinated and fluorinated compounds, aromatics, esters, and aldehydes (USEPA 1984). Absorption devices can be used separately or with other air pollution control equipment (e.g., to provide additional pollutant removal after incineration or condensation). Liquids are used as the absorbent; therefore, a media transfer of toxic pollutants can occur. Liquid-to-gas ratios, liquid temperature, and column height are also important parameters affecting efficiency.

In the absence of process specific data, absorption can be assumed to have maximum DREs in excess of 90 percent for VOC inlet concentrations over 250 ppmv, over 95 percent for VOC inlet concentrations over 500 ppmv, and over 98 percent for VOC inlet concentrations over 5,000 ppmv (USEPA 1984).

4. Condensation

Condensation is used as a control technique for some organic compounds. It cools the gas stream and transforms the gaseous compound to a liquid. Like absorption, condensation is one of the primary techniques used for product recovery; however, it is also used as an air-pollution control. Control of storage and process emissions is a common application. Condensers are frequently

used in series with other control equipment, including absorbers, incinerators, and adsorbers.

In the absence of process specific data, condensation can be assumed to have maximum DREs in excess of 50 percent for VOC inlet concentrations over 500 ppmv, over 80 percent for VOC inlet concentrations over 2,500 ppmv, and over 95 percent for VOC inlet concentrations over 5,000 ppmv (USEPA 1984).

5. Cyclones/Mechanical Collectors

Cyclones are seldom used as the sole primary means of particulate collection, but they often serve as "first stage" air-cleaning devices that are followed by other method of particle collection. Cyclone collection efficiency is probably more susceptible to changes in particulate characteristics (i.e., process variation) than are other types of devices. Therefore, care should be taken in the use of design efficiency to estimate actual operating conditions. Although very little compound-specific collection data are available, cyclone operation depends on physical parameters (particle size, density, velocity) as opposed to the chemical nature or properties of the material being collected. Thus, within reason, it may be possible to obtain and transfer efficiency data from known applications to unknown applications on processes with physically similar particulate and gas flows.

Cyclones are good as precleaners removing large particles because of their inability to capture particles smaller than 5 microns. Removal efficiencies range between 80 and 99 percent in a conventional cyclone for particles greater than 15 microns. A high-efficiency cyclone will capture particles greater than 5 microns (Stern 1977).

6. Fabric Filters

When properly designed and operated, fabric filters or baghouses are efficient collection devices, even for small particles. Vendor information is often a good source of collection efficiency information, as most units are designed for specific applications. As is true of cyclones, fabric filter performance is affected by process variations that affect the gas stream and by other variables such as

temperature and gas dew point. The particle collection mechanisms of these filters (like those of cyclones) usually depend solely on physical as opposed to chemical properties. Thus, data from known applications may be transferable.

Baghouses can be used with heavy grain loadings, but should not be used for oily, hygroscopic, or explosive dusts. Fabric filters can collect particle sizes ranging from submicron to several hundred microns in diameter. Removal efficiencies are generally greater than 99 percent for particles greater than 1 micron (Perry and Green 1984). Design parameters which affect the removal efficiency include the air-to-cloth ratio, the inlet particulate concentration, the temperature of the air, and the physical characteristics of the solids being removed.

7. Electrostatic Precipitators

Electrostatic precipitators (ESPs) remove from gas streams particles that have been electrically charged. They are not used to collect organic solids because of combustibility potential. Efficiency data are limited except for ESPs applied to combustion processes. The collection efficiency of an ESP depends on the physical characteristics of the particulate and the gas stream, and on the electrical resistivity of the pollutant to be collected. Electrical resistivity, in turn, can be affected by temperature, which may vary in some processes.

Theoretically, there is no minimum limit to the size of the particles that can be collected by an ESP. High-efficiency ESPs have efficiencies greater than 90 percent for particles in the 0.1- to 10.0-micron range (Stern 1977). However, the collection plates must be periodically cleaned in order to maintain a high overall efficiency, since the resistivity of the dust cake affects the voltage and corona current, and thereby the performance of the precipitator.

8. Wet Scrubbers

Wet scrubbers are used to collect organic and inorganic particulate matter and reactive gases. Scrubbers which often use water as the scrubbing medium, have the inherent potential of creating releases in the liquid medium. Like some other particulate collection equipment devices, scrubber designs are based on

physical parameters, so available efficiency data may be transferrable. The key factors in scrubber performance are particle size and scrubber pressure drop. As shown in Figure 7-1 for a venturi-type scrubber, a high particle removal efficiency can be achieved for larger particles and at higher pressure drops across the device (USEPA 1986d). The removal efficiency of a scrubber in removing gases or vapors depends on both the solubility of the gas or vapor in the liquid and the degree of saturation. Removal efficiency also depends on the contact time between the contaminate in the gas phase and the surfaces of the liquid phase.

C. Liquid and Solid Waste Control

Liquid (nonaqueous) and solid waste (including sludges and slurries) may be reduced by incineration, chemical treatment, physical treatment, recovery/reuse, and solidification/stabilization.

1. Incineration

Incineration is one of the most widely used technologies for hazardous waste disposal. Industrial wastes consisting of a PMN chemical fraction are often incinerated. These wastes include solvent cleaning/degreasing wastes, distillation and reactor bottoms, separator/detergent sludges, skimmer refuse, waste oils, polymer wastes, paint/ink sludges, thinners, pesticides/herbicides/insecticides, filter cakes, and contaminated fiber drums/boxes.

The quantity of chemical or other chemical of interest released to air from incineration of waste (containing the chemical) can be estimated using the Destruction and Removal Efficiency (DRE) of the incineration device. The DRE combines the efficiencies of both the destruction of the organic chemical of interest in the combustion chamber and, if applicable, the efficiency of the subsequent removal of any residual material from the stack effluent using control devices.

There are four types of incineration devices: incinerators, flares, industrial boilers, and process furnaces. For each of these four types of incineration there are many variations; however, all incineration can be grouped into one of the four types.

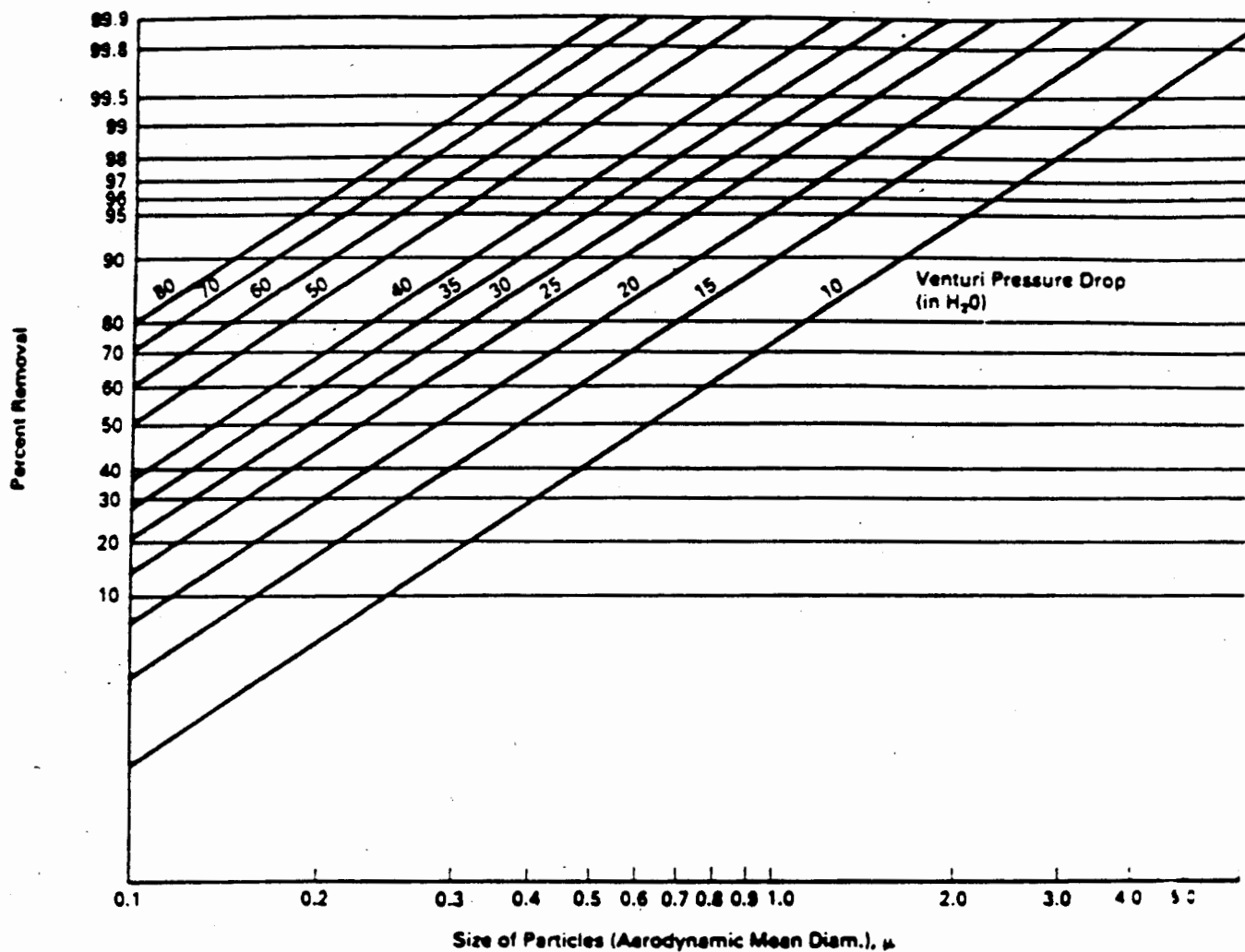


Figure 7-1. Venturi scrubber collection efficiencies (USEPA 1986d).

Incinerators are designed for the destruction of materials at high temperatures. Incinerators may be thermal or catalytic. Thermal incinerators rely on high temperatures, sufficient residence time, and adequate turbulence to insure high destruction efficiencies. Catalytic incinerators operate at somewhat lower temperatures as a catalyst promotes the oxidation. Although most VOCs are rapidly oxidized at temperatures over 1400°F, some compounds (e.g., halogenated hydrocarbons) require higher temperatures. While destroying one air pollutant, incineration may create other pollutants that require further treatment for removal from flue gases. For example, an incinerator that effectively destroys trichloroethylene may create hydrogen chloride that must then be removed by flue gas scrubbing. Flares are used to destroy purged gaseous organic compounds when it is not economical to recover the heat value of the gases. Industrial boilers are designed to generate steam through combustion of fossil fuel and may use hazardous wastes as a supplementary fuel. Process furnaces are integral components of a manufacturing process and are used for the recovery of material or energy.

Typically, the concentration of organics in the incinerator stack gas is low, resulting in high DREs. The best basis for estimating efficiency is actual measurement, which in PMN scenarios is unlikely, or the use of data that reflect typical efficiency during similar operations.

EPA compiled efficiency data for three of the four types of incineration devices from trial burns and other performance tests (CEB 1989a). From these data, well-operated full scale incinerators averaged between 99.994 and 99.99997 percent DRE. Boiler data averaged between 99.98 to 99.999 percent DRE. Data compiled for process kilns ranged from an average of 99.2 to 99.998 percent DRE. The DRE averages are difficult to interpret for PMN chemicals since trial burns occur under steady-state conditions and the actual DRE is dependent on the concentration and chemical class. These averages may be used for well-operated incineration devices. A DRE of 99 percent is suggested for release estimation to represent a worst case estimate.

Although flaring is widely used, information on the DRE for flares is limited. A 98 percent DRE can be achieved for flares provided they operate under the conditions listed in Table 7-1.

TABLE 7-1. OPTIMAL OPERATING CONDITIONS FOR FLARES

Type of flare	Exit velocity, V (ft/sec)	Heating value of H_T of gas stream ^a (Btu/scf)
Steam-assisted	$V < 60$	$H_T \geq 300$
	$60 \leq V_{\max(1)}^b$	$300 < H_T < 1000$
	$V < 400$	$H_T > 1000$
Non-assisted	$V < 60$	$H_T \geq 200$
	$60 \leq V_{\max(1)}^b$	$200 < H_T < 1000$
	$V < 400$	$H_T > 1000$
Air-assisted	$V < V_{\max(2)}^c$	$H_T > 300$

^a Heating value of total gas stream (not just listed chemical).

^b $V_{\max(1)} = e [1.424 + 0.0118 (H_T)]$ or $\log V_{\max(1)} = 1.424 + 0.0018 H_T$

^c $V_{\max(2)} = 28.54 + 0.0087 H_T$

Source: CEB 1989a, 40 CFR 60.18 (July 1986)

H_T should be calculated at conditions of 25°C (77°F) and 1 atmosphere (14.7 psia). For information on measurement and calculation of operating exit velocity and heating value of the gas stream, consult 40 CFR 60.18 (July 1986). Flares with values of less than 300 Btu/scf (steam- or air-assisted flares) or 200 Btu/scf (nonassisted flares) may or may not achieve 98 percent destruction. For example, a steam-assisted flare burning a volatile organic compound could be considered to have a 98 percent efficiency for that compound if its exit velocity and Btu value of the gas stream were within one of the three operating conditions listed in

Table 7-1 for this type of flare. This would allow an estimate of the control efficiency in absence of other data for the compound.

To estimate release from an incineration device, the volume of waste containing the PMN chemical or chemical of interest must be determined. Some typical wastes include spent cleaning solvents, filter cakes, still bottoms, sludges, and drum residues. Estimates of waste volume of the material of interest must be obtained from the PMN submitter or from manufacturers of existing chemicals.

To estimate the amount of chemical emitted to air from incineration, volume of chemical fed to the incineration device should be adjusted with the expected efficiency. This method provides an order of magnitude estimate only.